

Reminders

Upcoming due dates

Wed Feb 18 Data checkpoint

Fri Feb 20 D6

Machine Learning

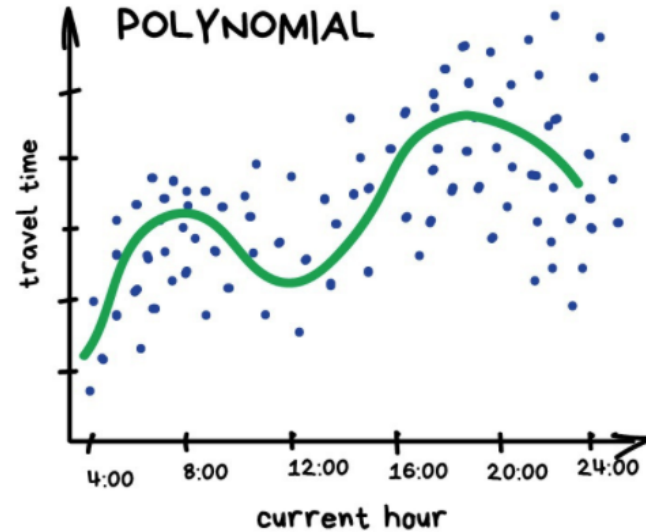
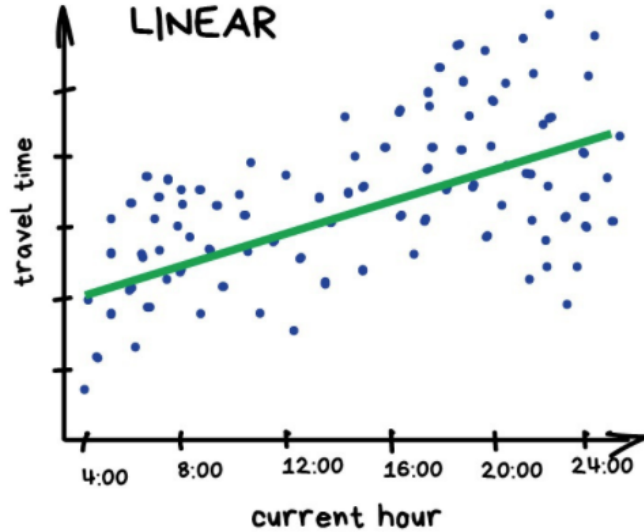
Data Science in Practice

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UC San Diego

PREDICT TRAFFIC JAMS



<https://forms.gle/HBiDbxR2oRBYxcUh7>

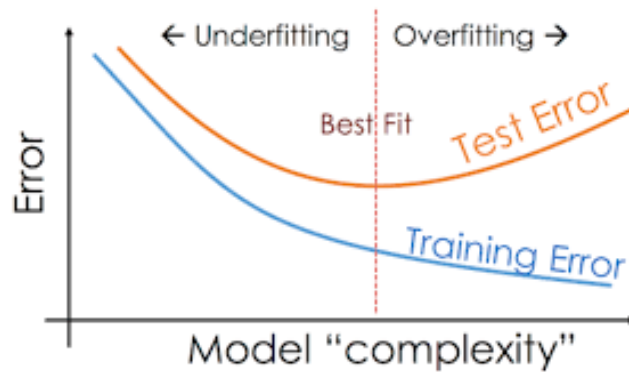
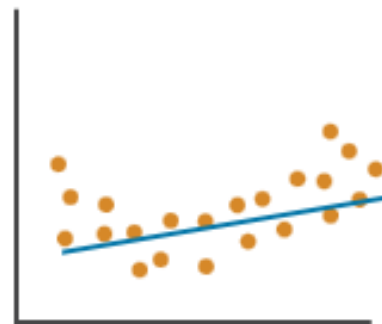
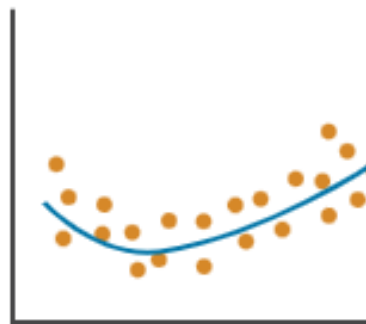
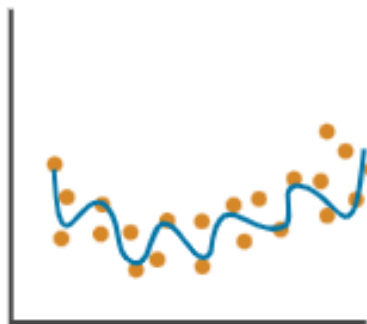


Overfit

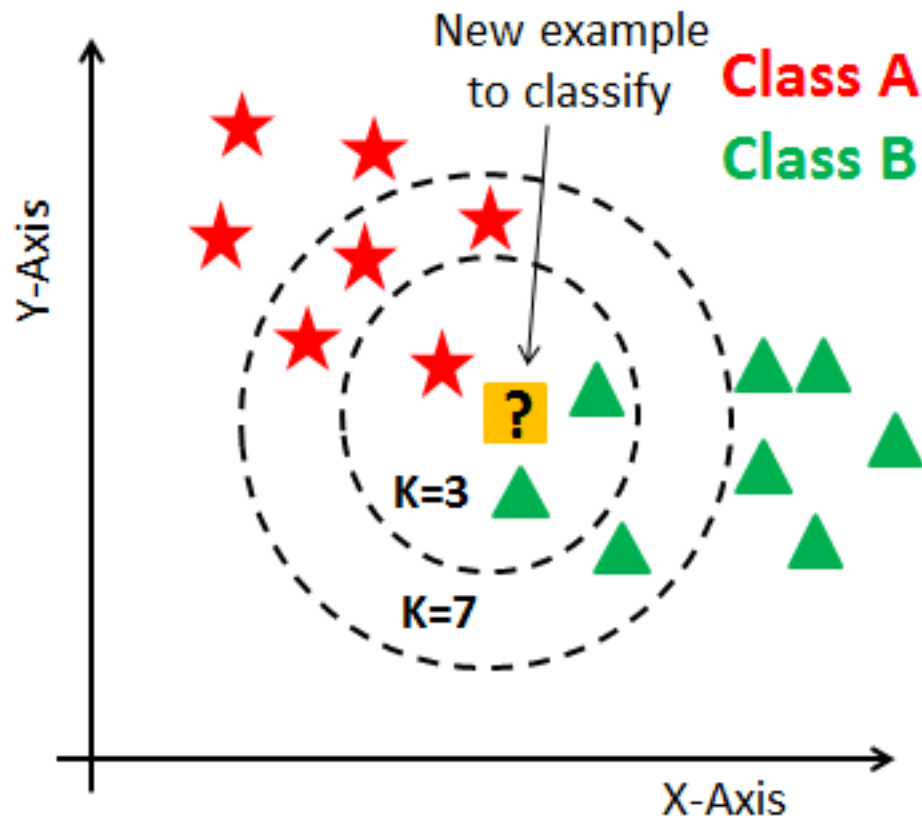
Best fit?

Underfit

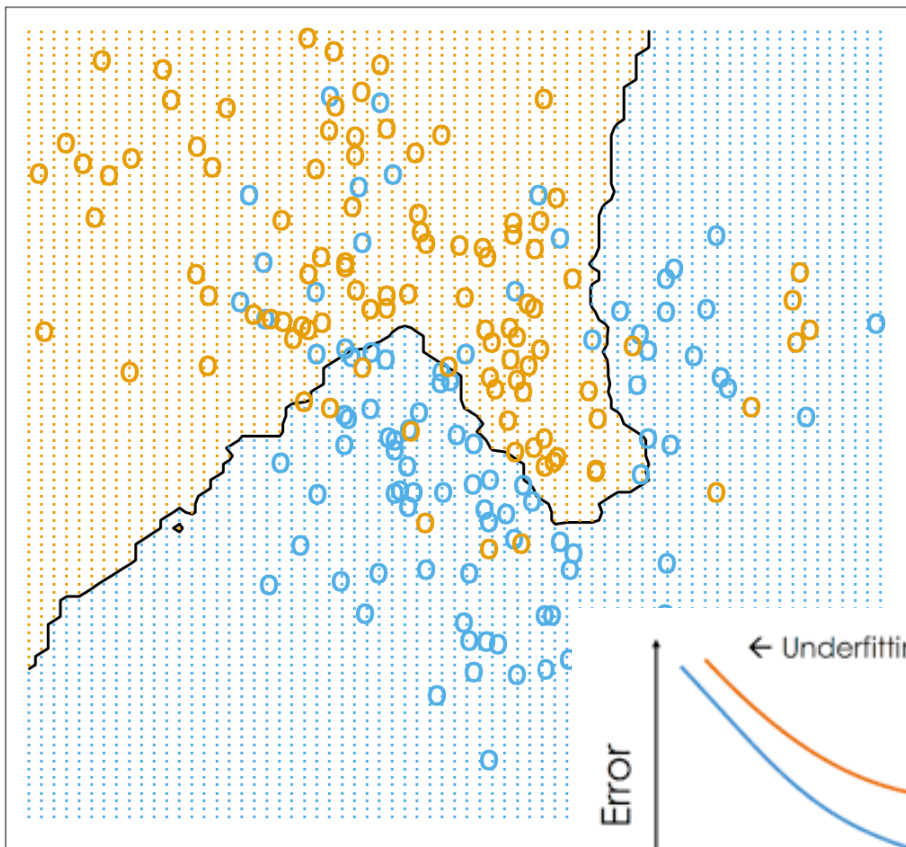
Regression



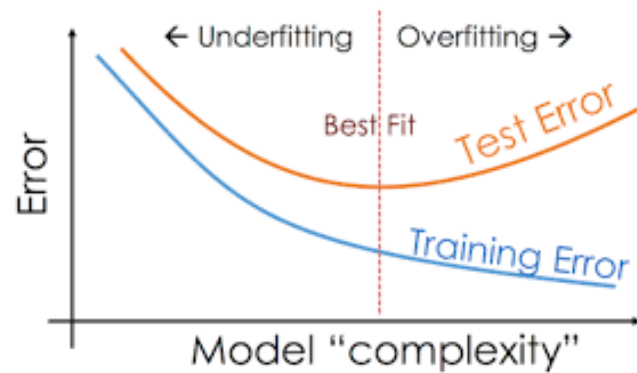
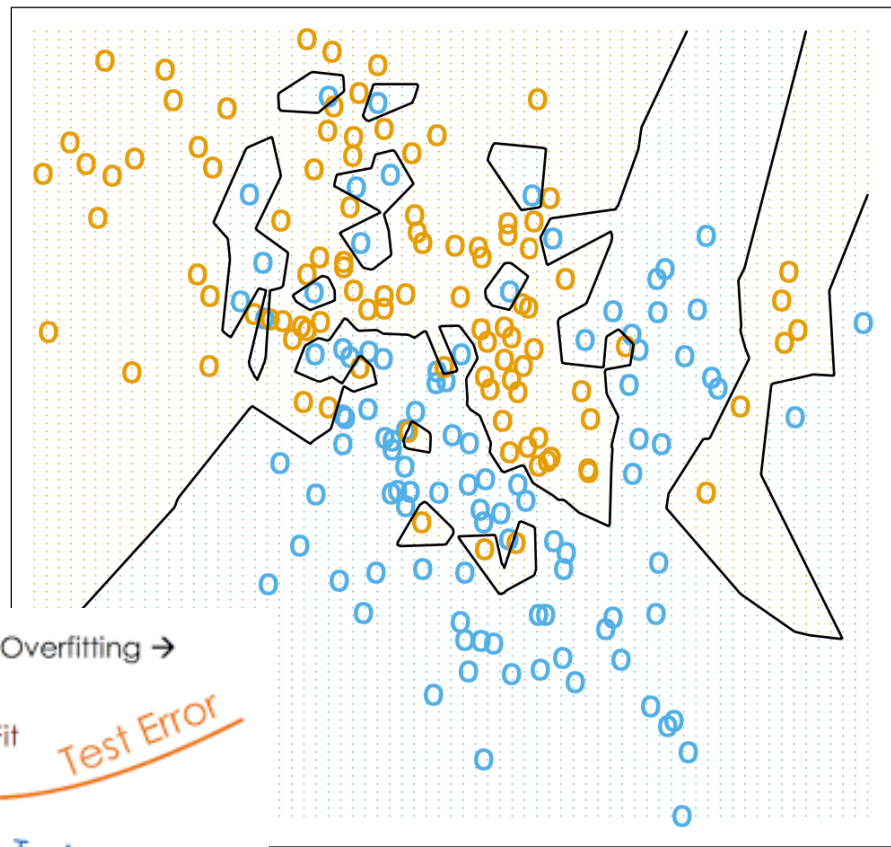
K-nearest neighbors



15-Nearest Neighbor Classifier

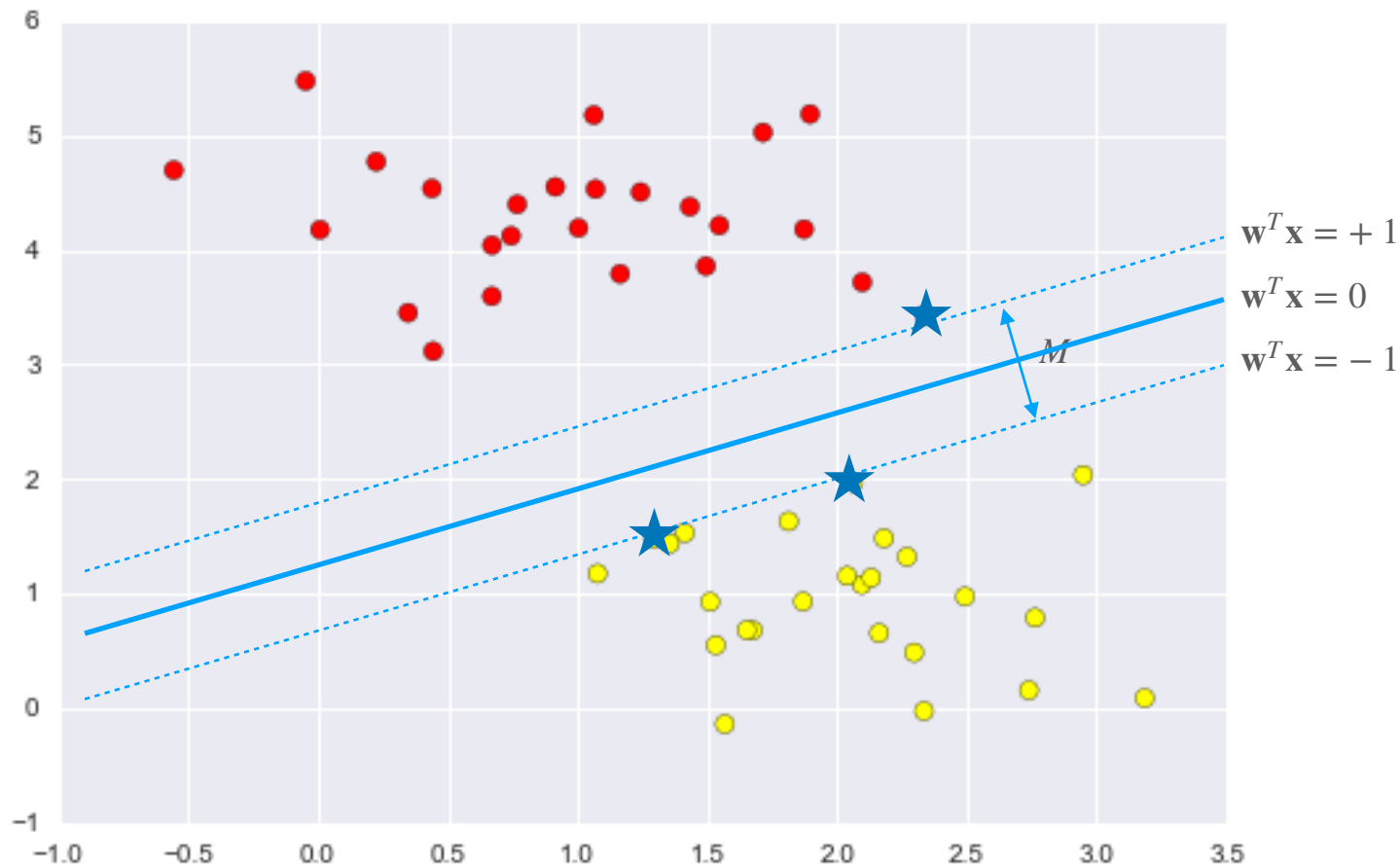


1-Nearest Neighbor Classifier



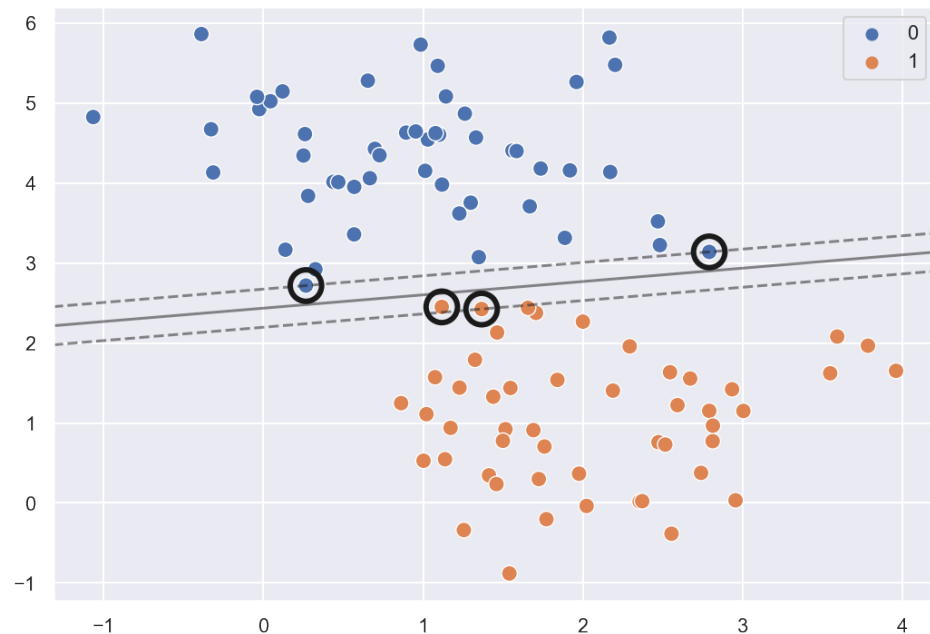
Support Vector Machines:

Find the support
data points that
maximize the
margin M



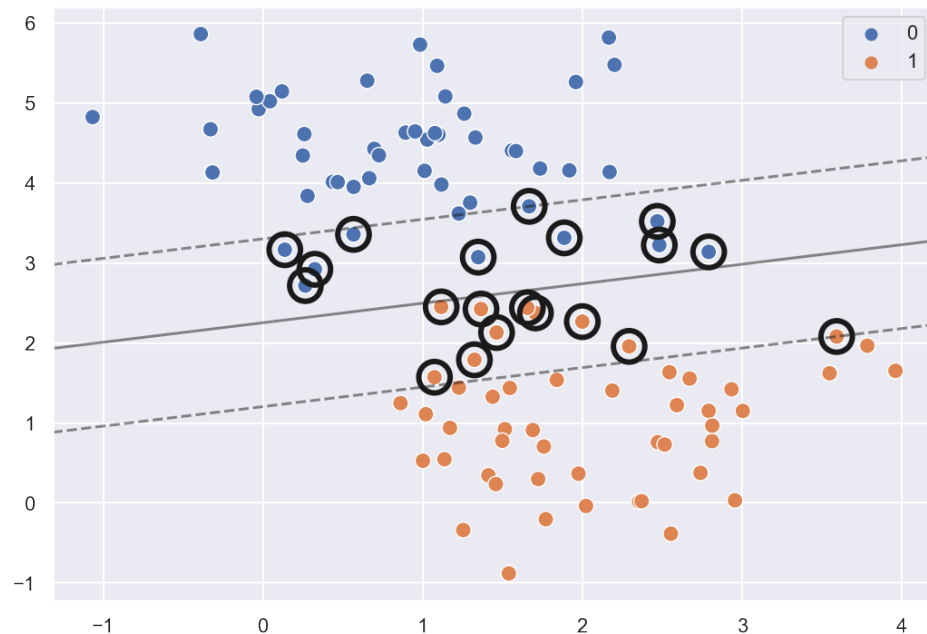
Hard margin

$C = 10.0$



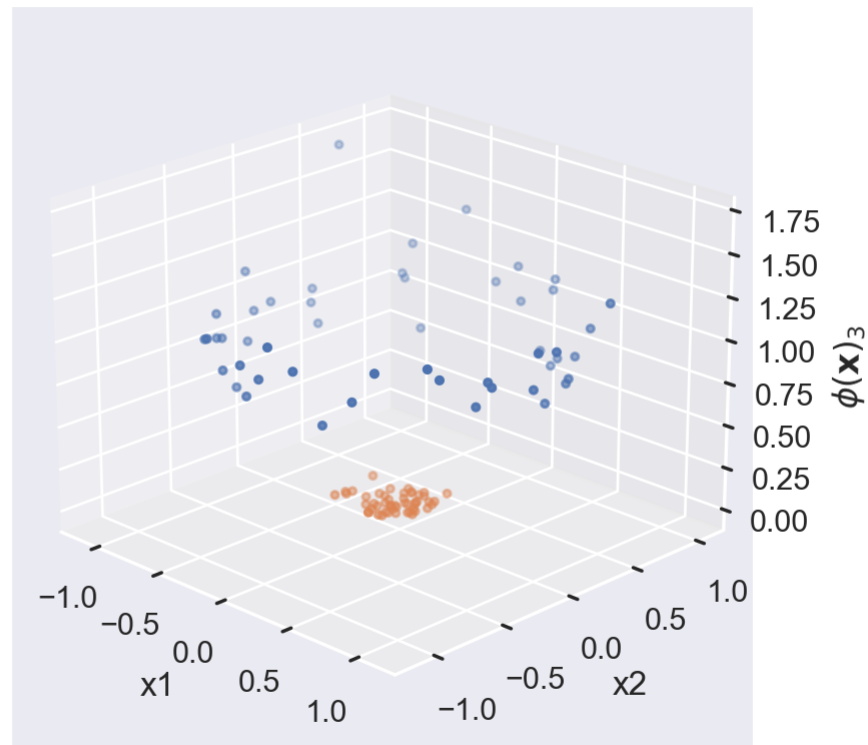
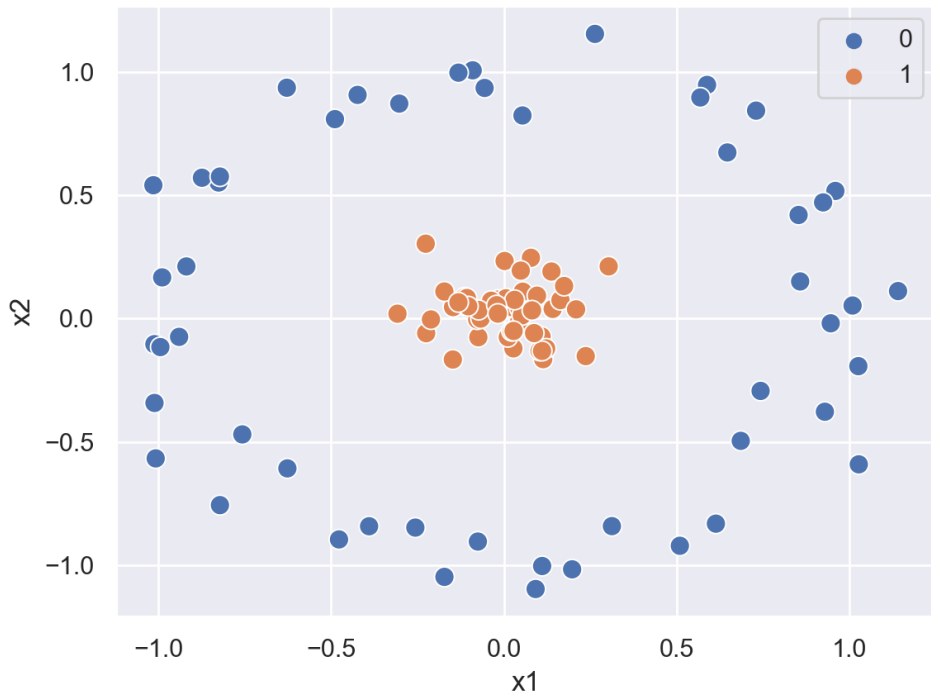
Soft margin

$C = 0.1$

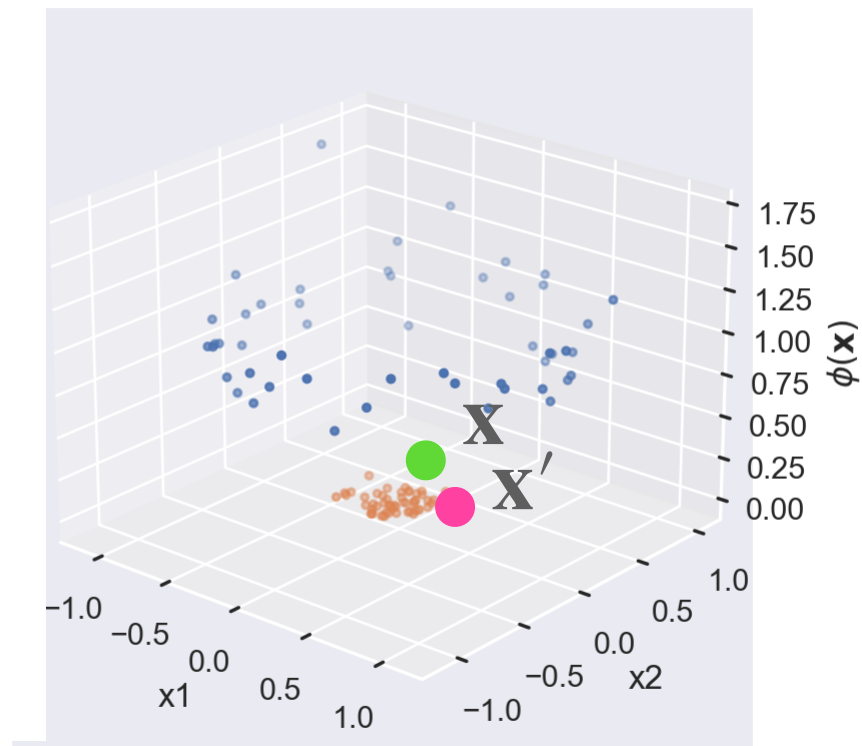
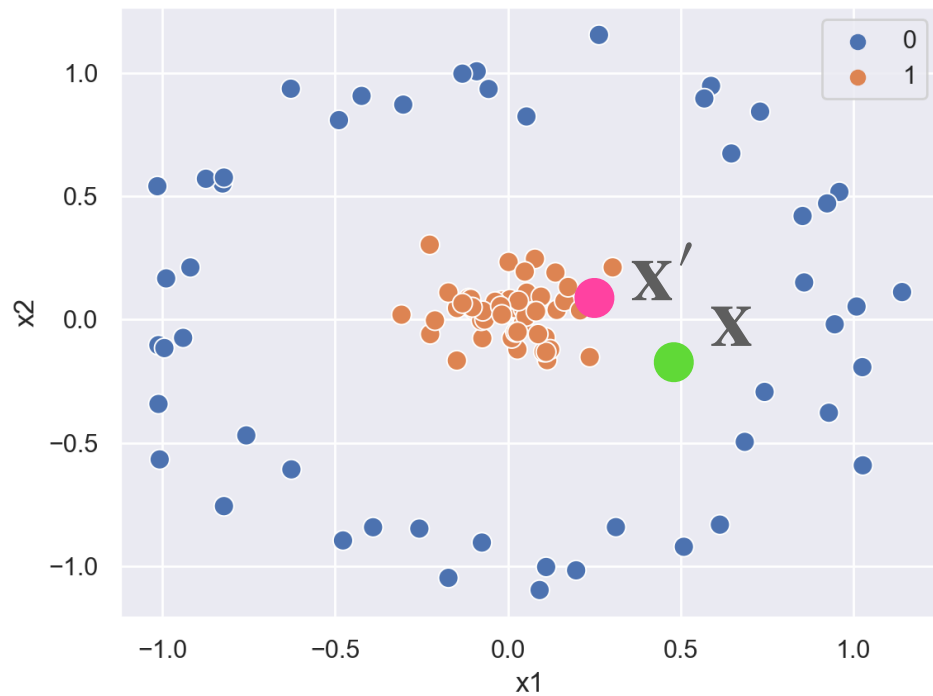


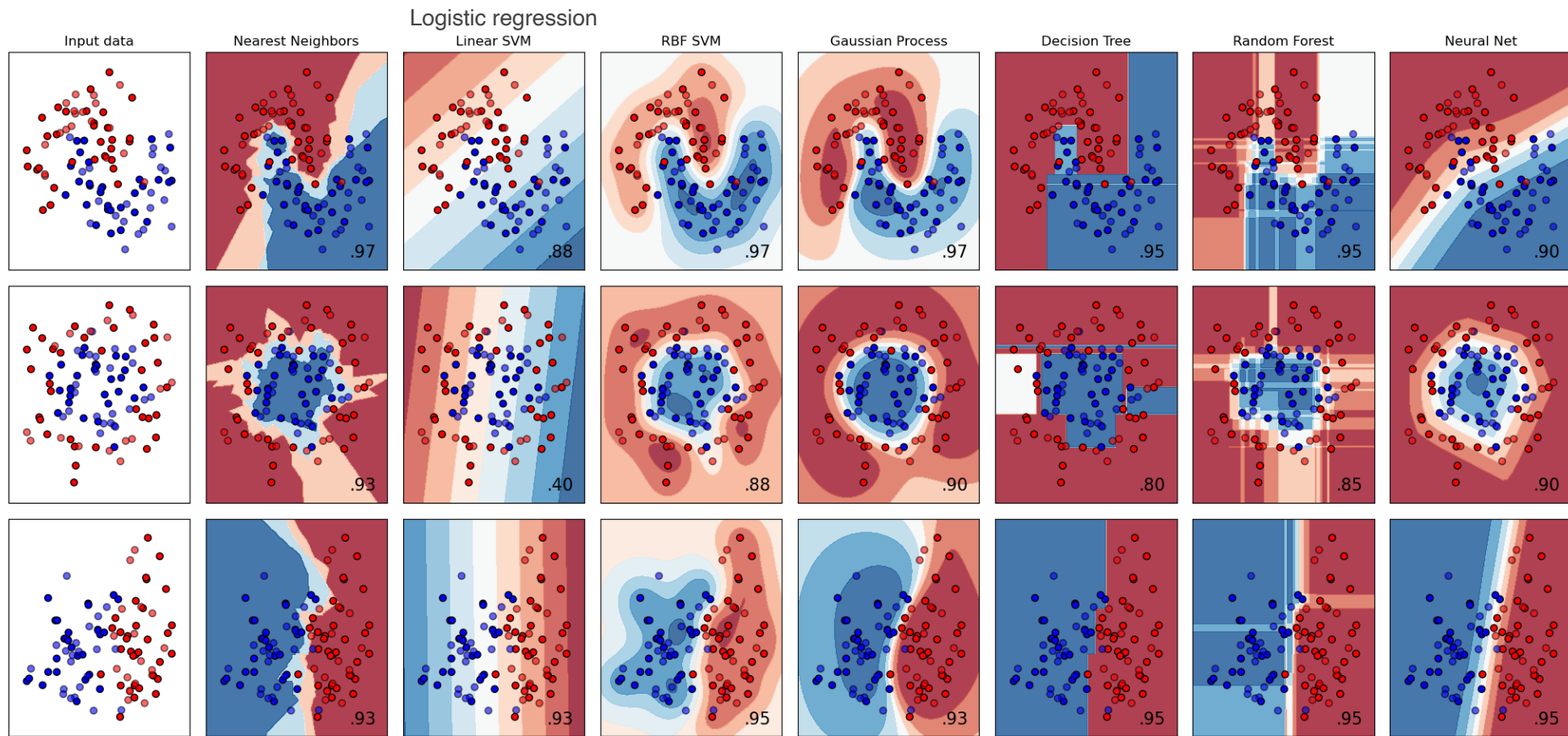
Find: $\arg \min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \times \sum_{i=1}^n (1 - y_i \times (\mathbf{w}^T \mathbf{x}_i + b))_+$

$$\phi(\mathbf{x}) = [x_1, x_2, x_1 \cdot x_2 + x_1^2 \cdot x_2^2]$$



More generally this is the polynomial kernel





https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html

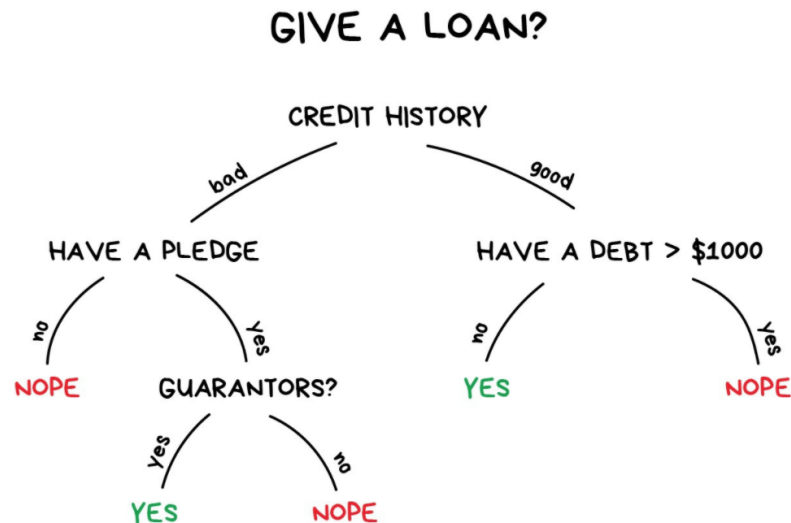
Linear vs Non-linear classifiers

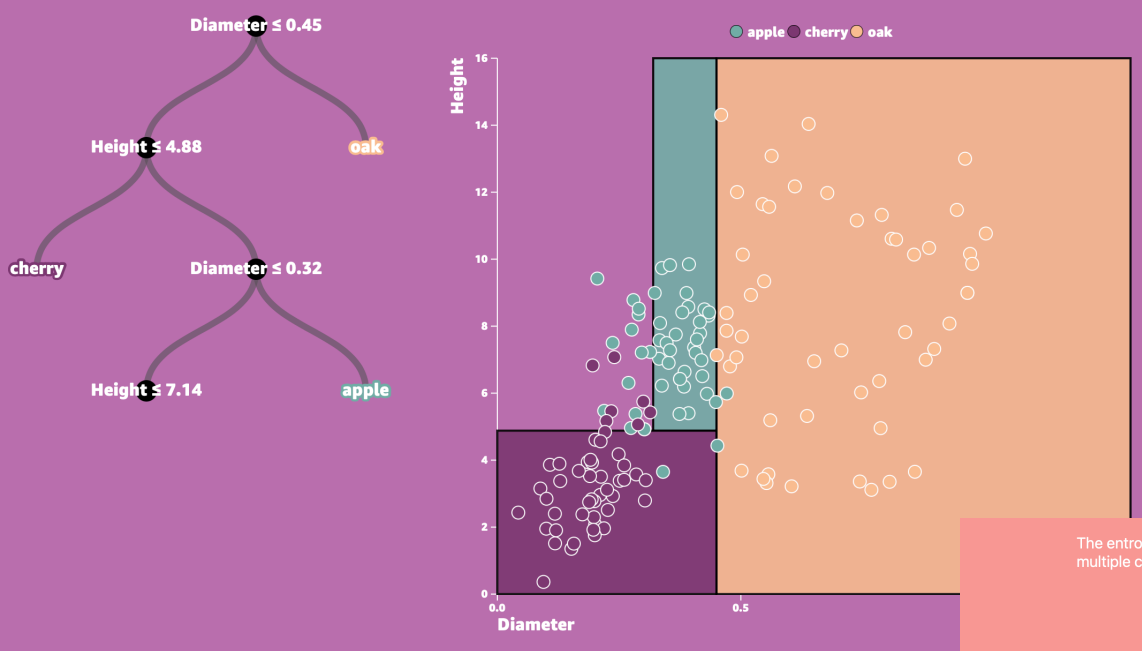
- LINEAR - Lower complexity, less overfitting, less power
 - Logistic regression
 - “Linear kernel” SVM
- NON-LINEAR - Higher complexity, more overfitting, more power
 - K-nearest Neighbors
 - “RBF kernel” SVM

Decision trees

ML that's human interpretable!

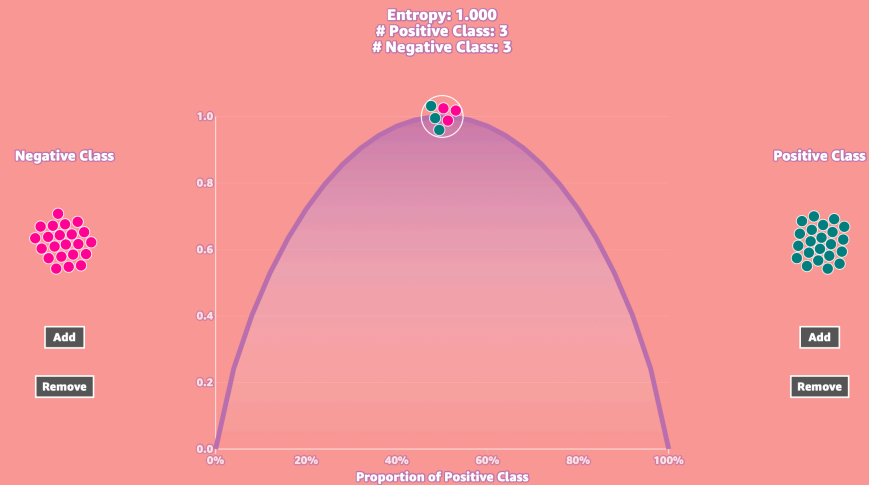
- Each node is
 - an if-then statement
 - visualizable as a decision boundary perpendicular to one axis in the data space
- Nodes arranged in a tree structure
- The axis and its threshold is determined by exhaustive search looking for the choice that minimizes the impurity of nodes after the split
- A tree can be seen as a collection of rules, it can be read by humans



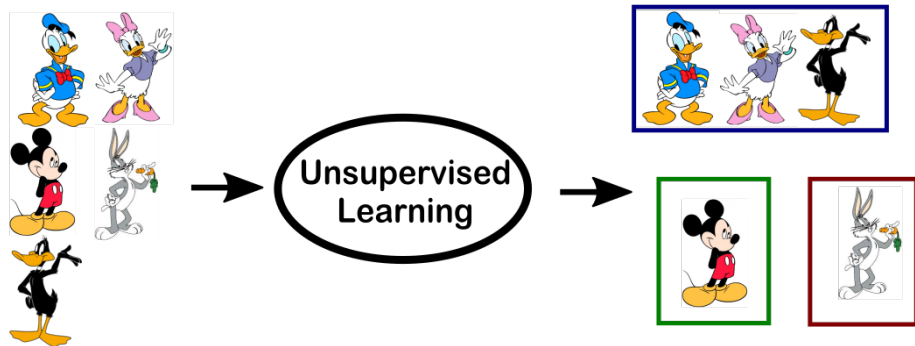


<https://mlu-explain.github.io/decision-tree/>

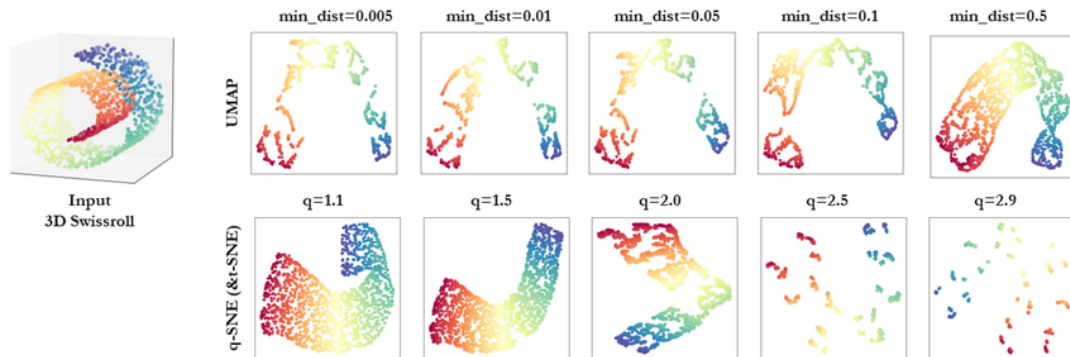
The entropy can be used to quantify the **impurity** of a collection of labeled data points: a node containing multiple classes is impure whereas a node including only one class is pure.



Unsupervised learning



Clustering:
categorical variables



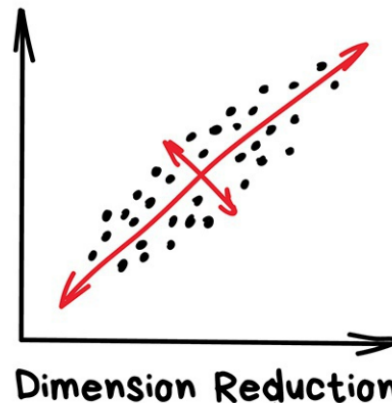
Dimensionality Reduction:
continuous variables

Dimensionality Reduction (Generalization)

"Assembles specific features into more high-level ones"

Nowadays is used for:

- Recommender systems (★)
- Beautiful visualizations
- Topic modeling and similar document search
- Fake image analysis
- Risk management



Popular algorithms: Principal Component Analysis (PCA), Singular Value Decomposition (SVD), Latent Dirichlet allocation (LDA), Latent Semantic Analysis (LSA, pLSA, GLSA), t-SNE (for visualization)

Modern dimensionality reduction algos
have hyper parameters that can make you think
patterns are there when they really aren't (and vv.)

<https://distill.pub/2016/misread-tsne/>

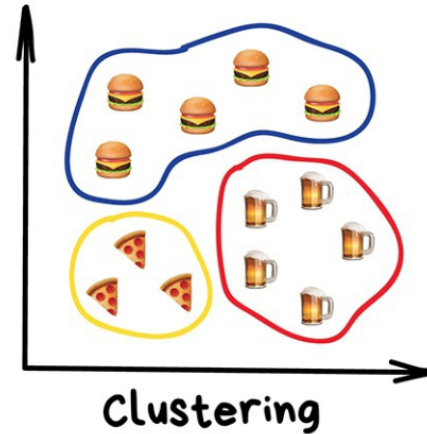
<https://pair-code.github.io/understanding-umap/>

Clustering

*"Divides objects based on unknown features.
Machine chooses the best way"*

Nowadays used:

- For market segmentation (types of customers, loyalty)
- To merge close points on a map
- For image compression
- To analyze and label new data
- To detect abnormal behavior



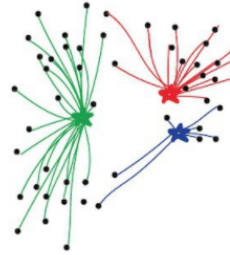
Popular algorithms: K-means_clustering, Mean-Shift, DBSCAN

PUT KEBAB KIOSKS IN THE OPTIMAL WAY

(also illustrating the K-means method)



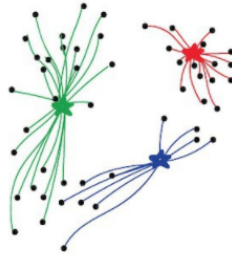
1. Put kebab kiosks in random places in city



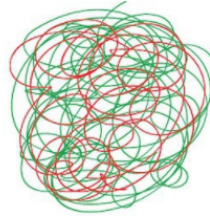
2. Watch how buyers choose the nearest one



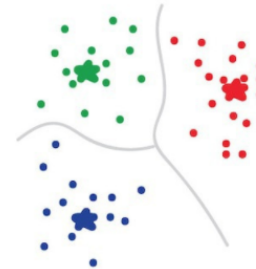
3. Move kiosks closer to the centers of their popularity



4. Watch and move again



5. Repeat a million times



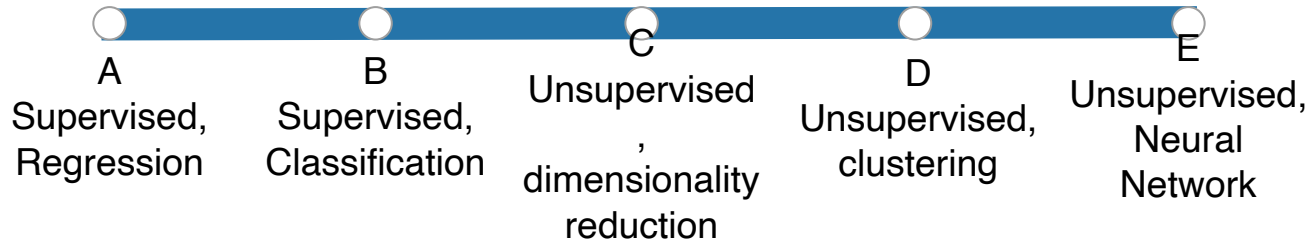
6. Done!
You're god of kebabs!



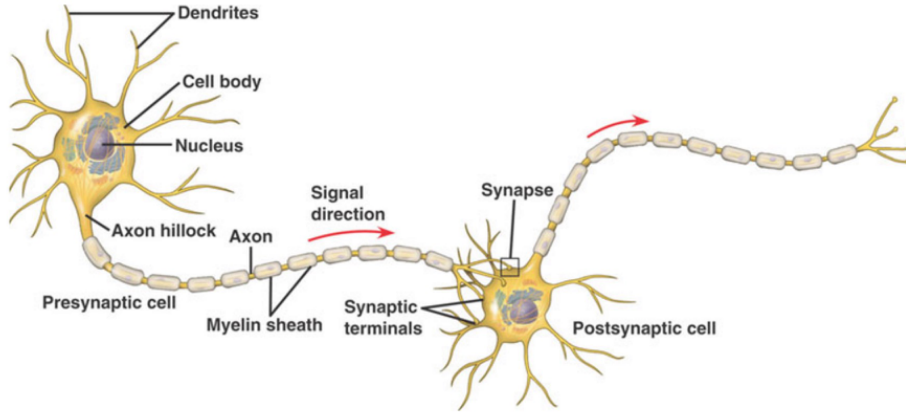
Prediction Approach

You want to predict someone's emotion based on an image.

How would you approach this with machine learning?

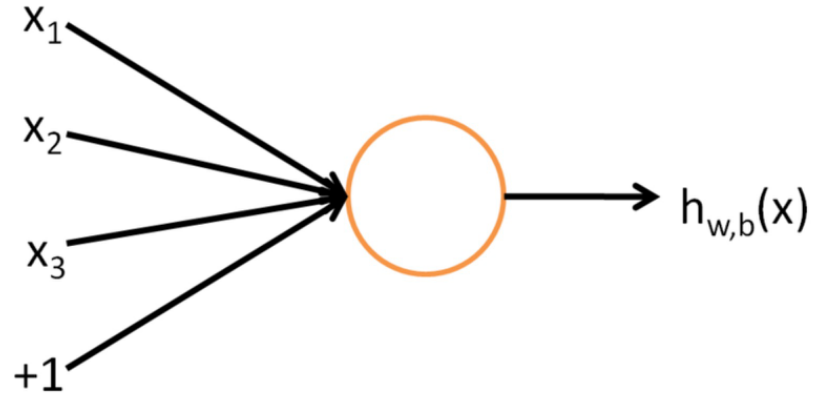


WHAT IS A NEURON?



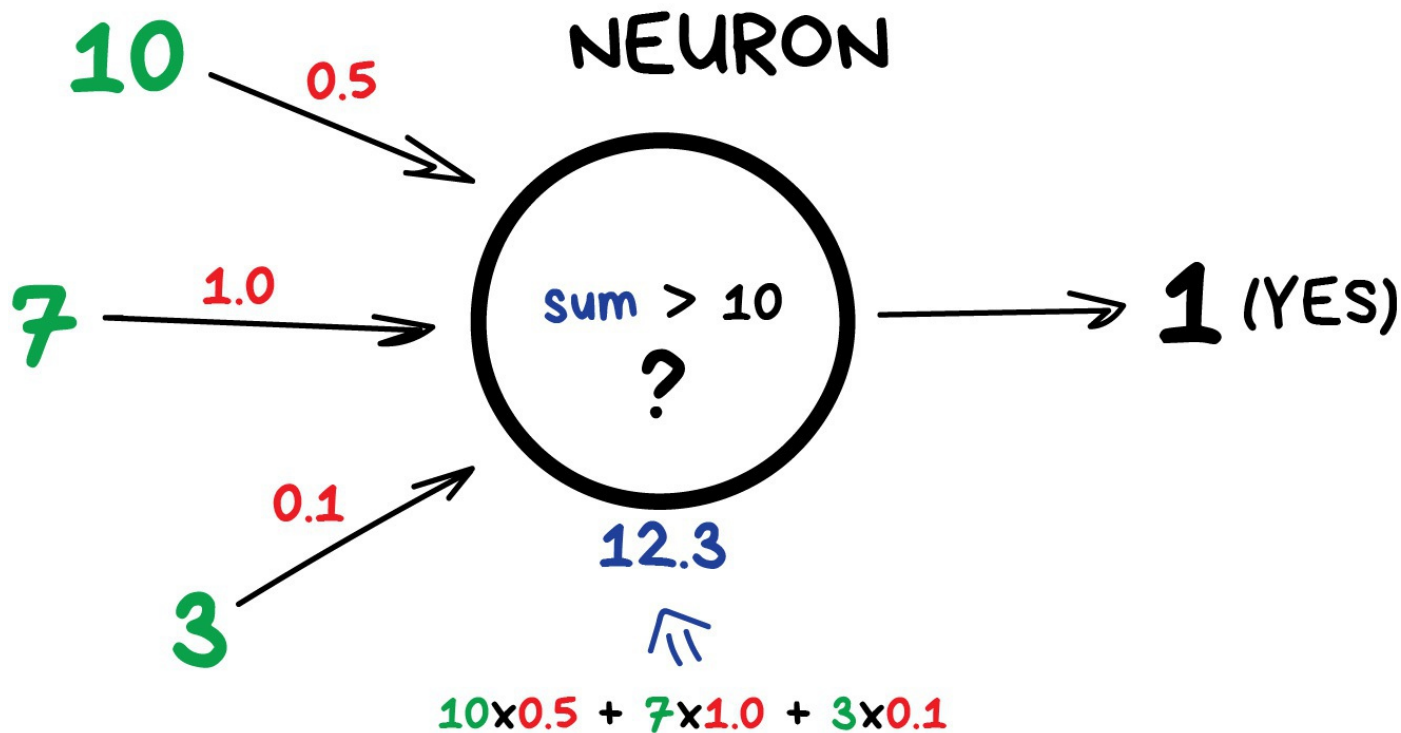
- Receives signal on synapse
- When trigger sends signal on axon

MATHEMATICAL NEURON

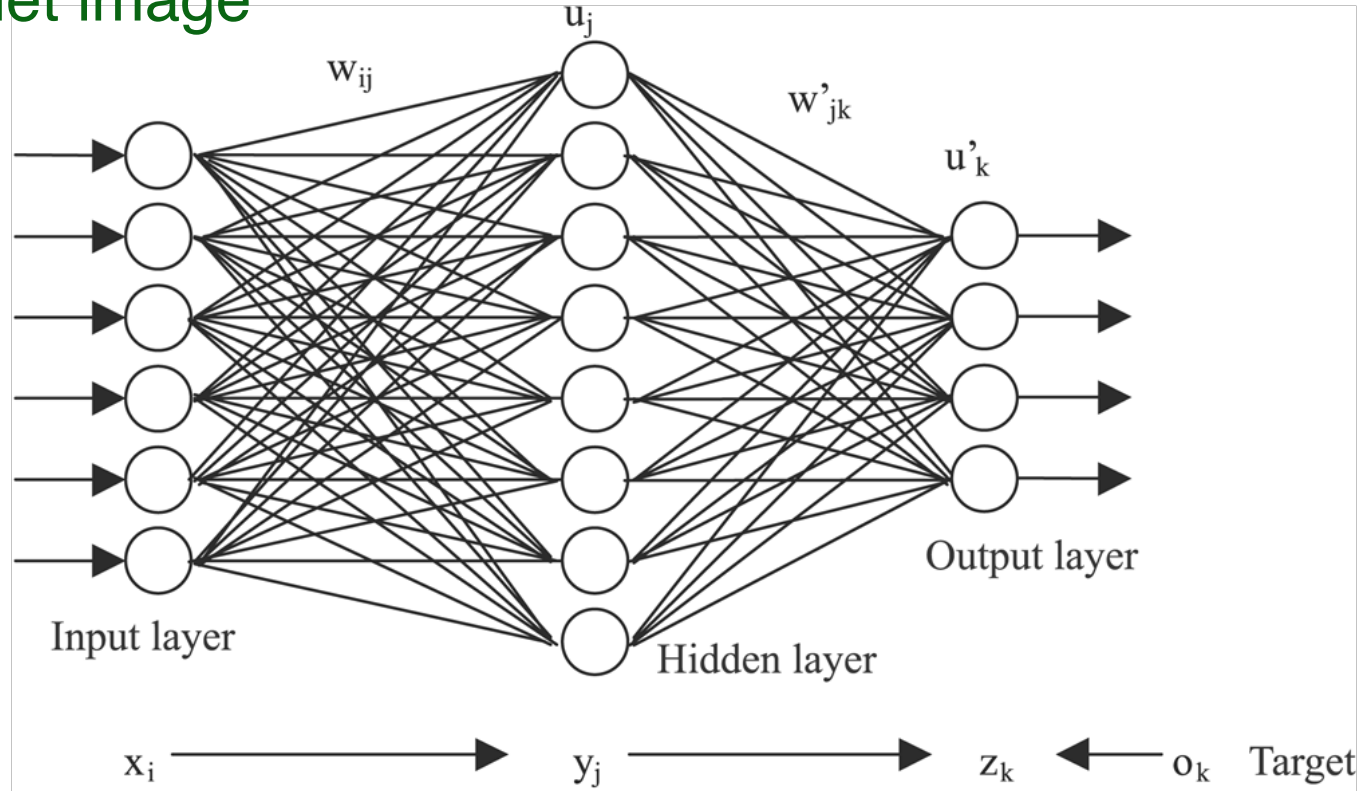


- Mathematical abstraction, inspired by biological neuron
- Either on or off based on sum of input

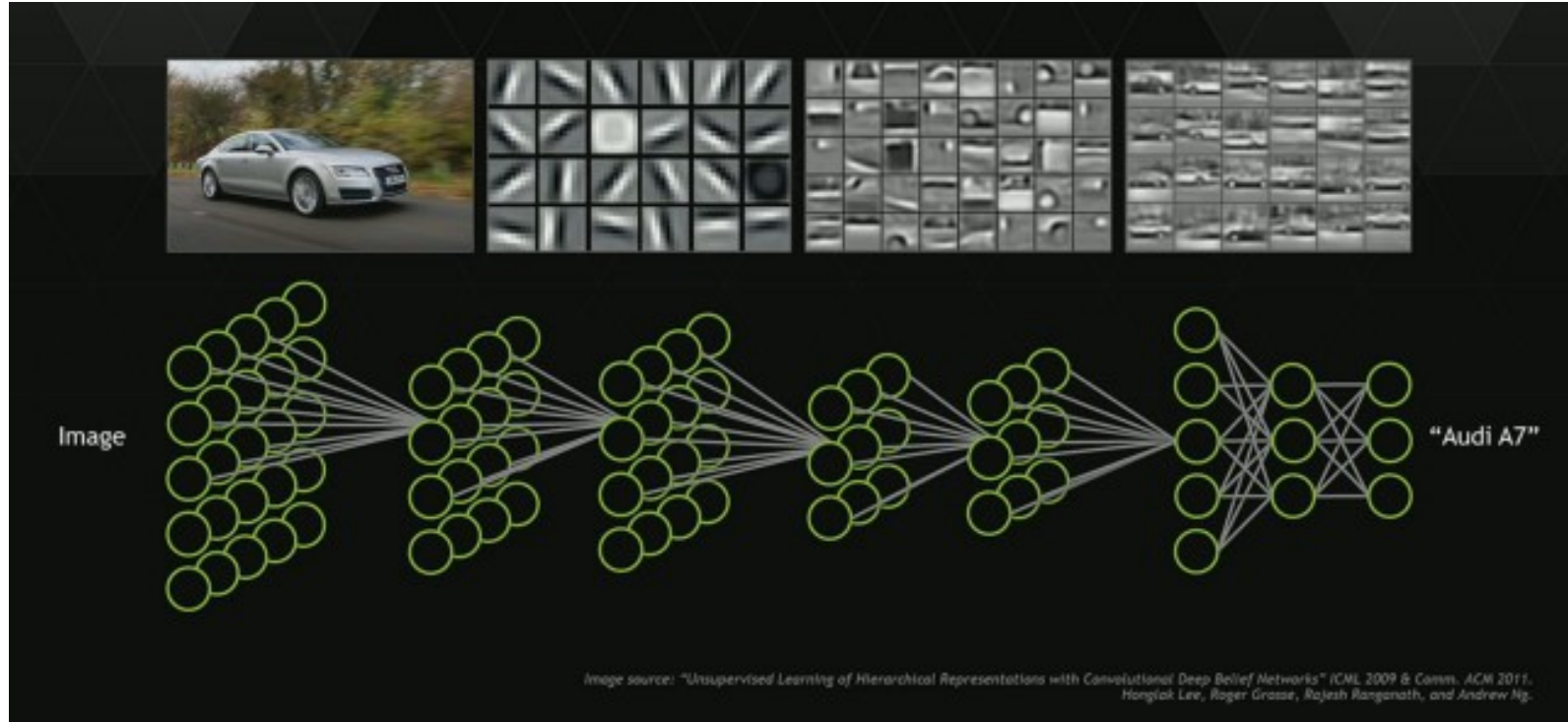
These weights tell the neuron to respond more to one input and less to another. Weights are adjusted when training — that's how the network learns. Basically, that's all there is to it.

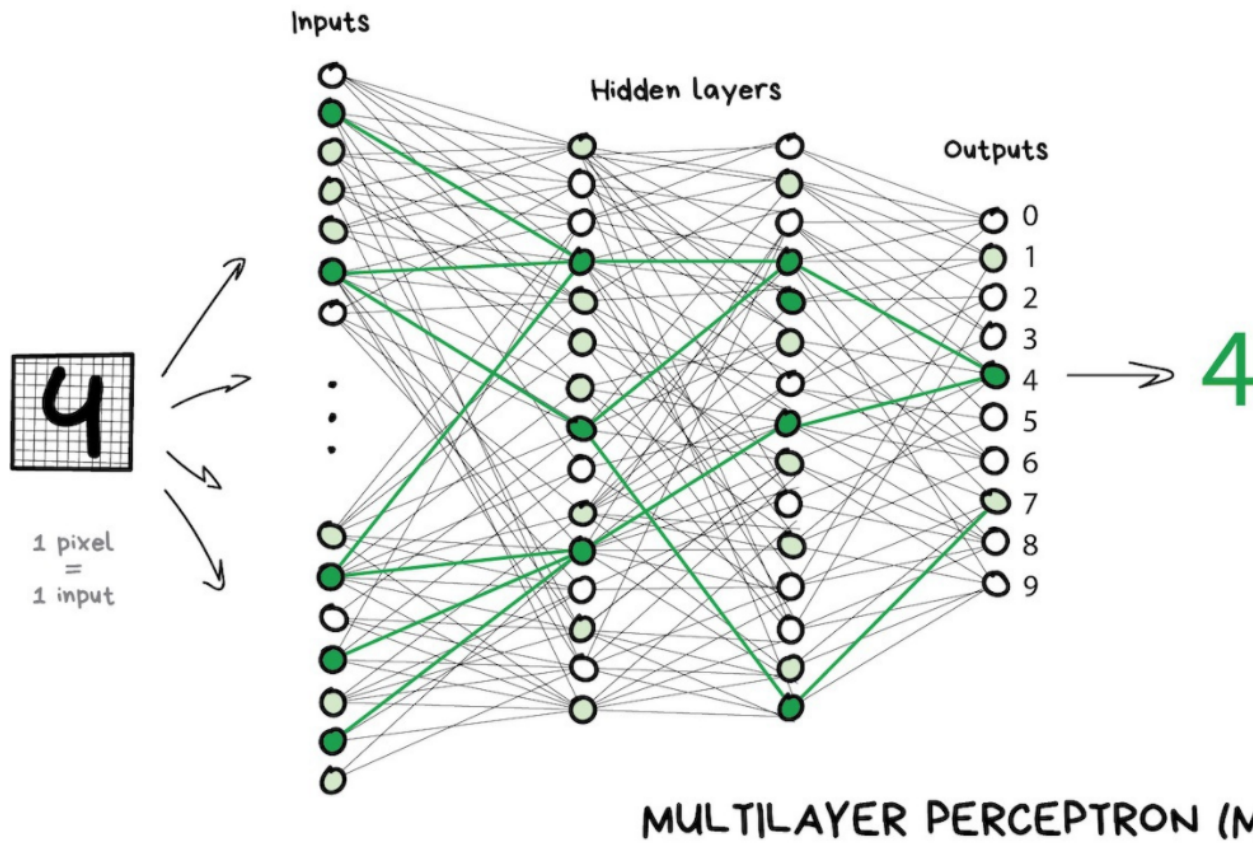


This will likely not be the last time you see this (mostly unhelpful)
neural net image

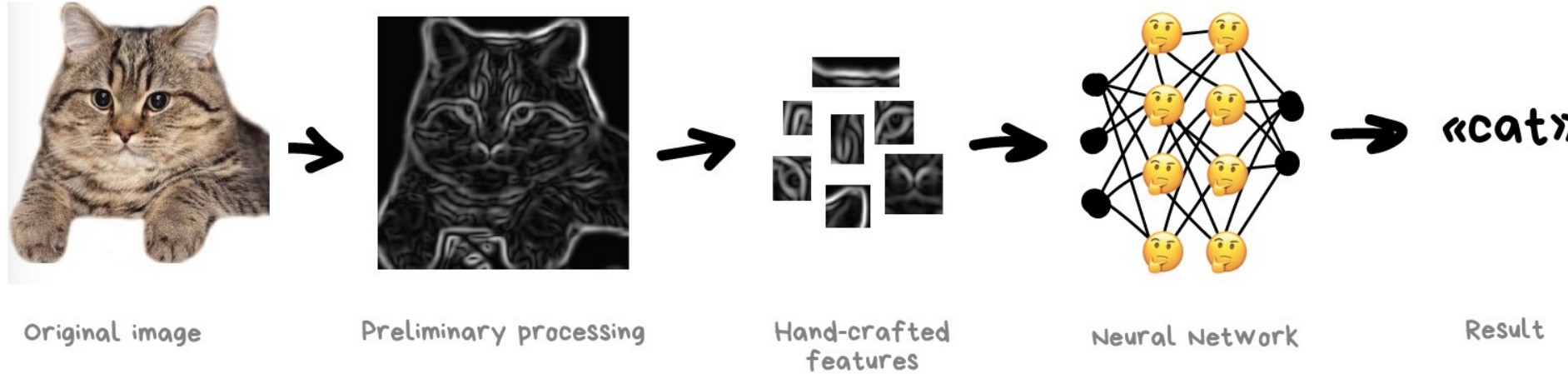


Convolutional neural networks

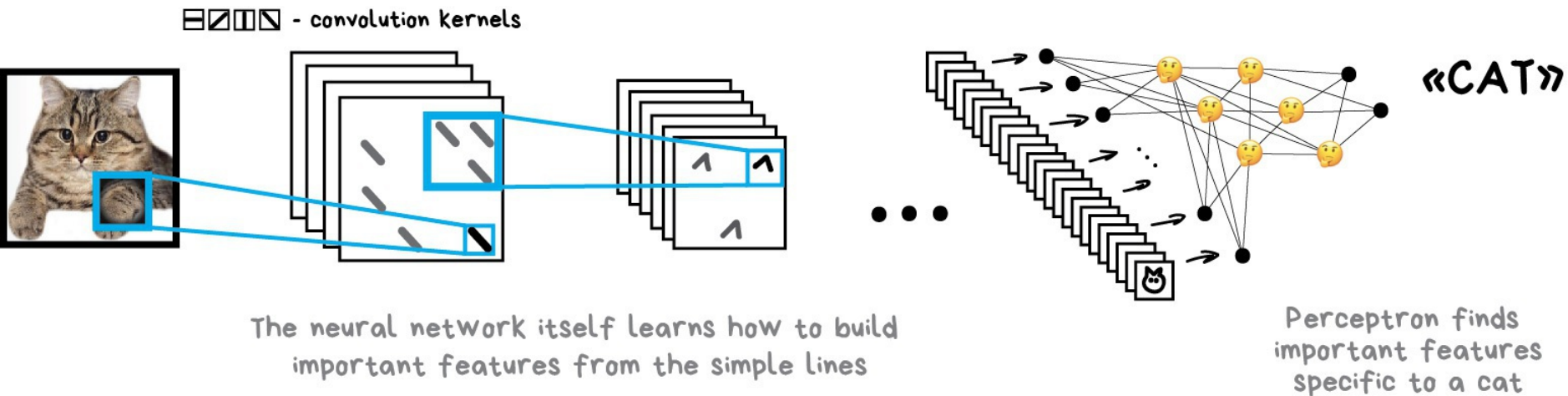




Manually labeling used to be the way...



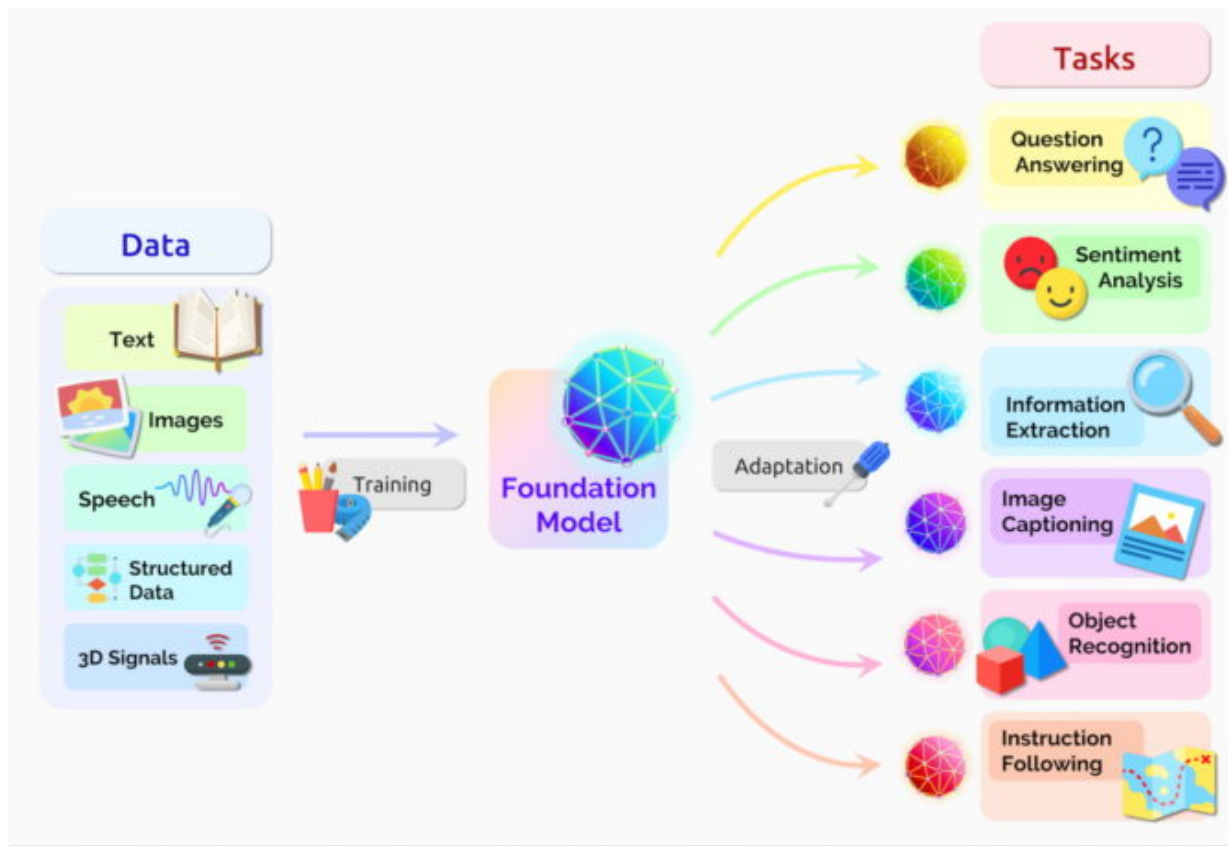
CNNs avoid manual features

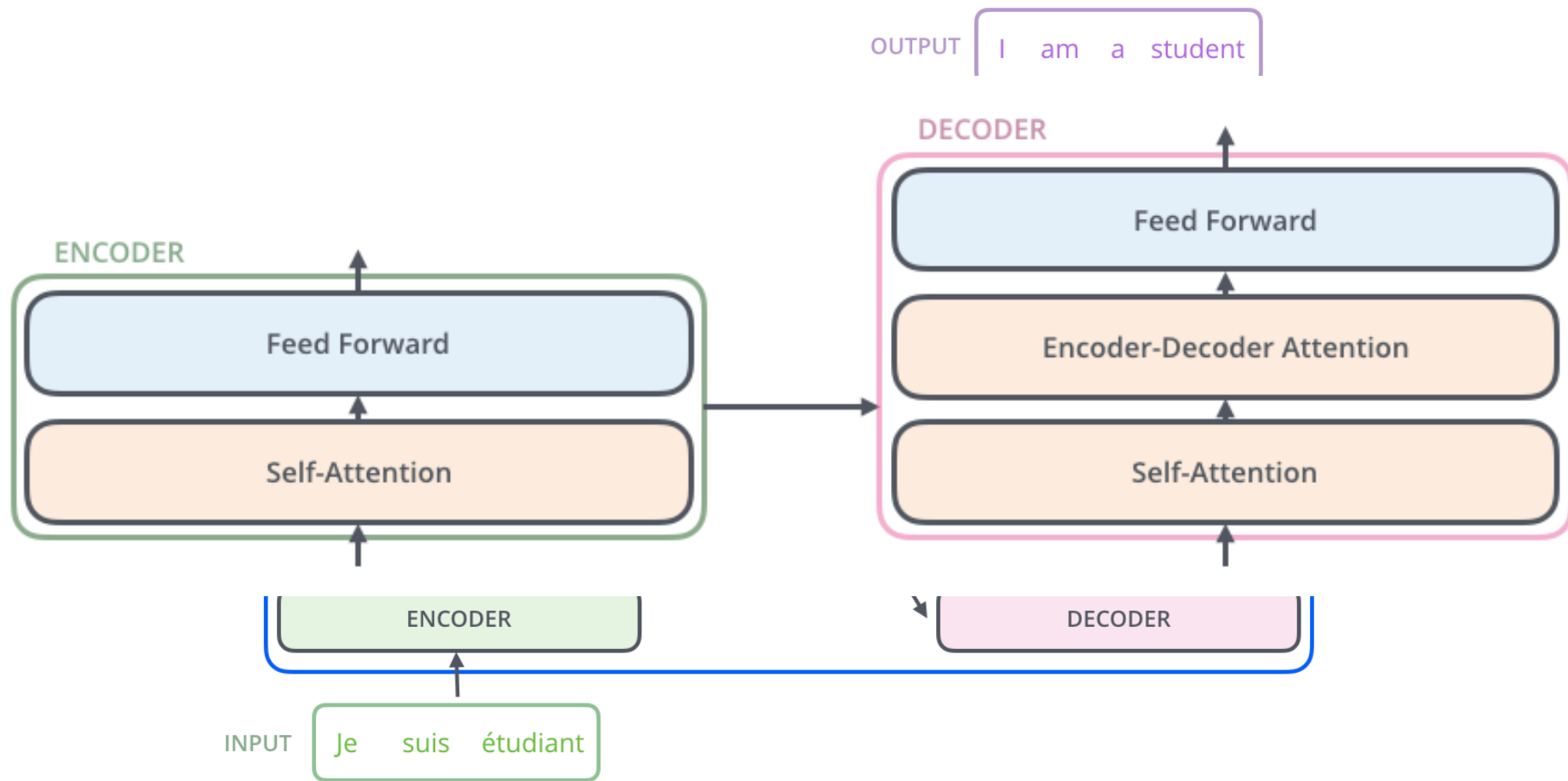


“CNNs are all the rage right now. They are used to search for objects on photos and in videos, face recognition, style transfer, generating and enhancing images, creating effects like slow-mo and improving image quality. Nowadays CNNs are used in all the cases that involve pictures and videos.”

CONVOLUTIONAL NEURAL NETWORK (CNN)







Much of DL success comes from semi-supervised tricks to avoid large hand labelled datasets

Masked LM

- **Solution:** Mask out $k\%$ of the input words, and then predict the masked words
 - We always use $k = 15\%$

the man went to the [MASK] to buy a [MASK] of milk

store gallon

↑ ↑

- Too little masking: Too expensive to train
- Too much masking: Not enough context

ML jargon

The model is the algorithm + hyperparameters; parameters are the result of training the model

- Algorithm:
 - some math to optimally solve a problem given some training data
- Solver:
 - The software implementation/approximation of the math; many different ways to get the job done
- Hyperparameters:
 - Knobs/Dials that affect how the algorithm solves the problem, the solver is one such
- Parameters:
 - the solution to the optimization problem being solved on the training data